LOS201 - Phạm Thế Vinh

Status	Finished
Started	Tuesday, 13 October 2026, 7:42 AM
Completed	Tuesday, 13 October 2026, 8:22 AM
Duration	39 mins 56 secs
Marks	30.00/40.00
Grade	**7.50** out of 10.00 (**75**%)

Question 1

Correct

Mark 1.00 out of 1.00

Remove flag

Which legacy bootloader uses Flash Image System partition tables in the chapter?

a. GRUB
b. coreboot
c. U-Boot
d. RedBoot ✓

Correct.

The legacy section associates RedBoot with FIS partition tables. Source: Chapter 10, slide 12.

The correct answer is: RedBoot

Question 2

Correct

Mark 1.00 out of 1.00

Flag question

After UBI attachment, which command mounts the named UBIFS volume?

a. mount -t proc ubi0:rootfs /mnt
b. mount -t ext4 /dev/mtd2 /mnt
c. mount -t squashfs ubi0:rootfs /mnt
d. mount -t ubifs ubi0:rootfs /mnt ✓

Correct.

The volume is mounted with filesystem type ubifs and the ubi0:rootfs source. Source: Chapter 10, slide 11.

The correct answer is: mount -t ubifs ubi0:rootfs /mnt

Question 3

Incorrect

Mark 0.00 out of 1.00

Remove flag

Which directory is assigned to volatile runtime storage in the example rootfs design?

a. /bin
b. /sbin ✗
c. /usr
d. /run

Incorrect.

The example mounts tmpfs at /run and /tmp for volatile data. Source: Chapter 06, slides 5, 6.

The correct answer is: /run

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

For kernel@100000 with reg = <0x00100000 0x00400000>, which value is its size in the shown layout?

a. 0x00100000
b. 0x00400000 ✓
c. 0x00500000
d. 0x00000000

Correct.

In the partition example, the second reg value is the partition size. Source: Chapter 10, slide 5.

The correct answer is: 0x00400000

Question 5

Quiz navigation

1 ✓ 2 ✓ 3 ✗ 4 ✓ 5 ✓ 6 ✓ 7 ✓ 8 ✗ 9 ✓
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Show one page at a time

Finish review

Correct

Mark 1.00 out of 1.00

Flag question

Which activity does source-code access directly enable?

a. Reviewing the program source implementation ✓

b. Setting documented program runtime options

c. Running the installed program executable

d. Reading the published program manual

Correct.

Access to source code lets developers study the implementation. Source: Chapter 01, slide 8.

The correct answer is: Reviewing the program source implementation

Question 6

Correct

Mark 1.00 out of 1.00

Flag question

Which top-level kernel directory contains architecture-specific code?

a. drivers

b. net

c. arch ✓

d. fs

Correct.

The kernel source diagram places architecture-specific code under arch. Source: Chapter 04, slide 9.

The correct answer is: arch

Question 7

Correct

Mark 1.00 out of 1.00

Flag question

A 32 MiB Flash has 1 MiB for the bootloader and 4 MiB for the kernel. How much remains for rootfs?

a. 28 MiB

b. 31 MiB

c. 24 MiB

d. 27 MiB ✓

Correct.

The remaining capacity is 32 - 1 - 4 = 27 MiB. Source: Chapter 10, slide 5.

The correct answer is: 27 MiB

Question 8

Incorrect

Mark 0.00 out of 1.00

Remove flag

Which division of responsibilities matches the UBI/UBIFS stack?

a. UBI manages Flash blocks; UBIFS manages files

b. UBIFS replaces the MTD driver; UBI replaces the VFS ✗

c. UBIFS manages UBI volumes; UBI manages filenames

d. UBI provides directory lookup; UBIFS provides device mapping

Incorrect.

UBI provides Flash-management services beneath the UBIFS filesystem. Source: Chapter 10, slide 8.

The correct answer is: UBI manages Flash blocks; UBIFS manages files

Question 9

Correct

Mark 1.00 out of 1.00

Flag question

Which U-Boot variable contains the Linux kernel command line?

a. bootdelay

b. fdt_addr_r

c. bootargs ✓

d. kernel_addr_r

Correct.

bootargs holds kernel command-line parameters. Source: Chapter 07, slide 9.

The correct answer is: bootargs

Question 10

Correct

Mark 1.00 out of 1.00

Flag question

In make -C $KDIR M=$(PWD) modules, what does -C $KDIR select?

- a. The application's installation prefix
- b. The kernel build directory ✓
- c. The module's runtime device node
- d. The target module load address

Correct.

The build command runs the kernel build system from KDIR. Source: Chapter 08, slide 7.

The correct answer is: The kernel build directory

Question 11

Incorrect

Mark 0.00 out of 1.00

Flag question

After configuration, which command builds BusyBox in the shown workflow?

- a. make clean
- b. make menuconfig ✗
- c. make
- d. make install

Incorrect.

The workflow uses make to build the BusyBox binary. Source: Chapter 11, slide 4.

The correct answer is: make

Question 12

Correct

Mark 1.00 out of 1.00

Flag question

Which location exposes device classes through sysfs?

- a. /proc/cpuinfo
- b. /etc/inittab
- c. /sys/class ✓
- d. /boot

Correct.

The slide lists /sys/class as part of the kernel's runtime device model. Source: Chapter 08, slide 13.

The correct answer is: /sys/class

Question 13

Correct

Mark 1.00 out of 1.00

Remove flag

Which early kernel object is associated with CPU setup and MMU enablement?

- a. sched.o
- b. main.o
- c. head.o ✓
- d. fork.o

Correct.

The flow-of-control slide assigns early CPU setup to head.o. Source: Chapter 05, slide 8.

The correct answer is: head.o

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

What does a storage partition represent?

- a. A running instance of an open file
- b. A device driver's callback structure
- c. A logical region of a storage device ✓
- d. A mapping from a filename to an inode

Correct.

Partitions divide storage into logical units. Source: Chapter 09, slide 3.

The correct answer is: A logical region of a storage device

Question 15

Correct

Which change can reduce the embedded kernel's size?

Mark 1.00 out of 1.00

Flag question

a. Disabling unused drivers and filesystems ✓
b. Including all optional debugging subsystems
c. Building support for unrelated board hardware
d. Enabling every available driver and filesystem

Correct.
Removing unused features reduces Flash and memory requirements. Source: Chapter 04, slide 13.
The correct answer is: Disabling unused drivers and filesystems

Question 16
Correct
Mark 1.00 out of 1.00
Flag question

Which command requests the current status of the sshd service?

a. sv down /etc/sv/sshd
b. sv up /etc/sv/sshd
c. sv restart /etc/sv/sshd
d. sv status /etc/sv/sshd ✓

Correct.
sv status queries the service state. Source: Chapter 11, slide 19.
The correct answer is: sv status /etc/sv/sshd

Question 17
Correct
Mark 1.00 out of 1.00
Flag question

Which named tool is associated with creating a cross-toolchain?

a. BusyBox init
b. GDB server
c. A serial terminal
d. Crosstool-NG ✓

Correct.
Crosstool-NG is listed among toolchain-related development tools. Source: Chapter 02, slide 14.
The correct answer is: Crosstool-NG

Question 18
Correct
Mark 1.00 out of 1.00
Flag question

Which kernel parameter specifies an alternative init program?

a. init= ✓
b. root=
c. console=
d. mem=

Correct.
init= can select an init program instead of the normal default. Source: Chapter 05, slide 10.
The correct answer is: init=

Question 19
Correct
Mark 1.00 out of 1.00
Flag question

Which Device Tree file form is normally passed to the kernel at boot?

a. A compiled .dtb blob ✓
b. A shared .dtsi include
c. The dtc compiler program
d. An uncompiled .dts source

Correct.
The kernel receives the compiled Device Tree blob. Source: Chapter 07, slides 13, 14.
The correct answer is: A compiled .dtb blob

Question 20
Correct
Mark 1.00 out of 1.00

Which build inputs are identified for an out-of-tree kernel module?

a. Kernel build headers and a module Makefile ✓
b. Only C-library headers and an application Makefile

Remove flag

c. Only rootfs contents and a filesystem image recipe

d. Only Device Tree sources and a bootloader script

Correct.

The slides specify kernel headers and a Makefile for module builds. Source: Chapter 08, slide 7.

The correct answer is: Kernel build headers and a module Makefile

Question 21

Incorrect

Mark 0.00 out of 1.00

Flag question

Why can tmpfs reduce Flash wear for temporary files?

a. Temporary writes increase Flash erase-block capacity

b. Temporary writes are always compressed into Flash

c. Temporary writes remove Flash bad-block requirements ⊗

d. Temporary writes are kept in volatile memory

Incorrect.

Using RAM for temporary files avoids corresponding Flash writes. Source: Chapter 06, slide 5.

The correct answer is: Temporary writes are kept in volatile memory

Question 22

Correct

Mark 1.00 out of 1.00

Flag question

Which selection factor concerns the readiness of compilers and related build tools?

a. Toolchain maturity ✓

b. Peripheral count

c. Board dimensions

d. Processor longevity

Correct.

Toolchain maturity concerns the quality and availability of the development tools. Source: Chapter 03, slide 12.

The correct answer is: Toolchain maturity

Question 23

Incorrect

Mark 0.00 out of 1.00

Flag question

Which command compiles board.dts into board.dtb?

a. dtc -I dtb -O dts -o board.dtb board.dts

b. dtc -I dts -O dtb -o board.dtb board.dts

c. dtc -I dtb -O dts -o board.dts board.dtb

d. dtc -I dts -O dtb -o board.dts board.dtb ⊗

Incorrect.

The input is DTS and the output is DTB in the compilation command. Source: Chapter 07, slide 15.

The correct answer is: dtc -I dts -O dtb -o board.dtb board.dts

Question 24

Incorrect

Mark 0.00 out of 1.00

Remove flag

When built-in initramfs contents change, which artifact normally needs rebuilding?

a. The Device Tree compiler

b. The kernel image

c. The hardware boot ROM

d. The bootloader environment ⊗

Incorrect.

The initramfs is embedded in the kernel image, so that image must be rebuilt. Source: Chapter 06, slide 13.

The correct answer is: The kernel image

Question 25

Correct

Mark 1.00 out of 1.00

Flag

Which command creates a SquashFS image?

a. mount

b. mksquashfs ✓

question

c. mkfs.ubifs

d. mkfs.ext4

Correct.

The image-creation example uses mksquashfs. Source: Chapter 09, slide 9.

The correct answer is: mksquashfs

Question 26

Correct

Mark 1.00 out of 1.00

Flag question

An embedded board has little RAM. Which goal directly addresses this limit?

a. Reducing application memory usage ✓

b. Reducing network transmission errors

c. Reducing Flash erase frequency

d. Reducing external connector count

Correct.

RAM is one of the constrained resources identified in the introduction. Source: Chapter 01, slide 3.

The correct answer is: Reducing application memory usage

Question 27

Correct

Mark 1.00 out of 1.00

Flag question

A portable Linux device drains its battery too quickly. Which mobile-Linux concern should the team examine first?

a. Power management ✓

b. Filesystem naming conventions

c. Software redistribution terms

d. Multimedia encoding format

Correct.

Power management is a stated focus of mobile Linux. Source: Chapter 01, slide 14.

The correct answer is: Power management

Question 28

Incorrect

Mark 0.00 out of 1.00

Remove flag

Which command family is appropriate for listing Linux network interfaces?

a. modinfo

b. ip

c. lsmod

d. depmod ✗

Incorrect.

The runtime-inspection slide names ip or ifconfig for network devices. Source: Chapter 08, slide 13.

The correct answer is: ip

Question 29

Correct

Mark 1.00 out of 1.00

Flag question

Which U-Boot variable identifies the RAM address used for the kernel?

a. initrd_addr_r

b. fdt_addr_r

c. kernel_addr_r ✓

d. bootdelay

Correct.

kernel_addr_r is the kernel load address variable in the table. Source: Chapter 07, slide 9.

The correct answer is: kernel_addr_r

Question 30

Incorrect

Mark 0.00 out of 1.00

Flag question

What does bootstrap code do for a compressed zImage?

a. Loads a desktop login session

b. Starts configured network services ✗

c. Mounts the user home filesystem

c. Mounts the user home filesystem

d. Decompresses the kernel payload

Incorrect.

The bootstrap code decompresses compressed kernel formats such as ARM32 zImage. Source: Chapter 05, slide 5.

The correct answer is: Decompresses the kernel payload

Question 31

Correct

Mark 1.00 out of 1.00

Flag question

Which Linux feature helps vendors use processors from different families?

a. Dependence on one proprietary instruction set

b. Support for multiple hardware architectures ✓

c. Support for one fixed processor architecture

d. Dependence on a single processor vendor

Correct.

Linux supports many hardware architectures. Source: Chapter 01, slide 5.

The correct answer is: Support for multiple hardware architectures

Question 32

Incorrect

Mark 0.00 out of 1.00

Flag question

A project comparison considers only OS royalties. Which important cost has been omitted?

a. The license renewal total already included in the comparison

b. The per-device license total already included in the comparison

c. Engineering and support costs over the product lifetime

d. The royalty total already included in the comparison ✗

Incorrect.

The chapter highlights development, integration, maintenance, and setup as costs beyond licensing. Source: Chapter 01, slides 5, 10.

The correct answer is: Engineering and support costs over the product lifetime

Question 33

Correct

Mark 1.00 out of 1.00

Flag question

Which utility creates the UBIFS filesystem image from rootfs content?

a. mount

b. ubinize

c. mkfs.ubifs ✓

d. ubiattach

Correct.

mkfs.ubifs creates the filesystem image; ubinize packages UBI images. Source: Chapter 10, slide 9.

The correct answer is: mkfs.ubifs

Question 34

Correct

Mark 1.00 out of 1.00

Flag question

Which pair belongs to the commercial embedded Linux group in the chapter?

a. Raspberry Pi OS and Ubuntu Core

b. Buildroot and OpenEmbedded

c. Yocto Project and Buildroot

d. Wind River and MontaVista ✓

Correct.

Wind River and MontaVista are listed as commercial embedded Linux providers. Source: Chapter 02, slide 15.

The correct answer is: Wind River and MontaVista

Question 35

Correct

Mark 1.00 out of 1.00

Which Device Tree property is used to match hardware to a driver?

a. reg

b. interrupts

Flag question

c. status

d. compatible

Correct.

The matching example uses compatible strings in the device node and driver table. Source: Chapter 08, slides 19, 20.

The correct answer is: compatible

Question 36

Incorrect

Mark 0.00 out of 1.00

Remove flag

What does MTD stand for?

a. Memory Translation Driver

b. Mapped Transfer Descriptor

c. Memory Technology Device

d. Managed Timing Device

Incorrect.

MTD is the Memory Technology Device subsystem. Source: Chapter 10, slides 2, 3.

The correct answer is: Memory Technology Device

Question 37

Correct

Mark 1.00 out of 1.00

Flag question

Which storage interface is associated with JFFS2?

a. SATA block storage

b. USB mass storage

c. MTD raw Flash

d. NVMe block storage

Correct.

JFFS2 is designed for raw Flash accessed through MTD. Source: Chapter 09, slide 8.

The correct answer is: MTD raw Flash

Question 38

Correct

Mark 1.00 out of 1.00

Flag question

Which optional script starts a separate logging process for a runit service?

a. finish

b. run

c. log/run

d. supervise/control

Correct.

The service-structure slide identifies log/run as the logging script. Source: Chapter 11, slide 15.

The correct answer is: log/run

Question 39

Correct

Mark 1.00 out of 1.00

Flag question

Which design choice best suits a resource-constrained embedded system?

a. Selecting components needed for its task

b. Keeping all debugging features in production

c. Installing every available software package

d. Reserving large buffers for unused functions

Correct.

Resource-constrained designs optimize software and hardware for the application. Source: Chapter 02, slide 4.

The correct answer is: Selecting components needed for its task

Question 40

Correct

Mark 1.00 out of 1.00

Flag

Which mount option allows a filesystem image file to be mounted through a loop device?

a. -o sync

b. -o noexec

question

c. -o ro

d. -o loop

Correct.

The example mounts rootfs.img using mount -o loop. Source: Chapter 09, slide 15.

The correct answer is: -o loop

Finish review